



IE 211 MATHEMATICAL MODELLING

Abdullah Gul University



Inventory Management in a Drilling Machine Company

ERD Drilling Machines, Ankara

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INTRODUCTION

ERD Drilling Machines, established in 2016 in Ankara, points out as an important company within Turkey's domestic production landscape. Experts in the production of ground drilling machines, underground drilling equipment, ground survey drilling, and water drilling machines, ERD is a frontrunner in providing these tools within the country. Notably, they also offer spare parts for their range of products. Among their array of offerings, the ZT-250 and Extreme x6 models shine out with their efficiency and reliability. These are some products which are produce in ERD Drilling machines:



Figure 1. ZT-250



Figure 2. Extreme x6

ERD sets a due date of 4 to 6 weeks for these highly sought-after products. In this process, ERD Drilling Machines have 13 workers in the factory, they maintain a balanced workforce with 9 blue-collar workers and 4 white-collar workers. Their team involves a general manager, sales representative, accountant, and automotive engineer, each playing a special role in the company's operations. Despite their success, ERD encounters persistent challenges in production planning and stock management. These issues have directly led to delays in meeting product due dates, impressing their operational efficiency. In our project, we're searching the root causes of these delays, and reviewing the intricacies of ERD's production planning and stock management processes. Our aim is to organize operations and advance the company's ability to meet deadlines effectively, by approaching these issues directly.

LITERATURE REVIEW

As a result of the literature review, some studies examining the effect of inventory management on company performance were found and these studies were briefly mentioned.

In their study, Obermaier and Donhauser (2012) investigated the effect of stocks held by companies on company performance. The study showed that low-performing companies carry the lowest inventory, while high-performing companies have the highest inventory.

The study conducted by Victoria and Ukpere (2014) examined the relationship between inventory management and organizational performance in 7 Up bottling companies in Nile Mile Enugu. According to the research results, flexibility in stock control management has been determined as an important strategy in achieving corporate performance. Companies have benefited significantly from inventory control management through factors such as easy storage and recall of materials, increasing sales effectiveness and reducing operating costs. The research also emphasized that effective stock control management is considered one of the skills that any organization must acquire.

Chalotra's research (2013) examined the role of inventory management to promote small business growth. It was conducted on 44 manufacturing companies and states that proper inventory management leads to an increase in the economy. Inventory management can grow the economy by enabling businesses to use their resources effectively. The study emphasizes that good inventory management increases competitiveness. Businesses can gain competitive advantage by providing flexibility in meeting demands. According to Chalotra's findings, proper inventory management reduces costs. By effectively managing their inventory, businesses can keep costs under control and reduce waste.

Shin and team (2015) examined the relationship between financial statement data and inventory management efficiency in their study on manufacturing companies in the USA. One of the main findings of the study is that there is a positive relationship between profitability and inventory management efficiency. In other words, it has been stated that more effective inventory management practices are associated with company profitability. Shin and team emphasized that the effect of inventory efficiency on profitability is significantly important according to firm size. Based on the time period and measurement types, the study concludes that more efficient inventory management practices result in higher firm profitability.

PROBLEM DEFINITION

The company's inefficient production scheduling process is causing production delays, customer dissatisfaction, and declining profitability.

The production of ZT250 - full hydraulic drilling machine should be done in 4 weeks on average according to Schelling planning, but the time extends to 6 weeks on average weeks due to the inefficient manufacturing scheduling process. This is mainly caused by inadequate stock quantities.

The roots of this problem, which can be called insufficient stock management, lie behind two possible related situations. Firstly, they don't collect data and plan the production management process according to this data. Lack of professionalism is the first point of this case which can be seen as a mistake. Even in the amateur business, collecting the data and using the scarce sources in the consideration of data is the first step that should be taken into account in terms of the development of the company. The second problem is the lack of area and financial resources. When we are dealing with an insufficient number of stocks of particular components, these two deficiencies are the most frequent problems, and it is also the case in our problem.

We have asked our company supervisor to collect some useful information. After nearly one month we have received this data and built a mathematical model with this data.

- ❖ The company has 80 m^3 stock space for the materials which are used in the production of ZT-250.
- ❖ The company wants to purchase materials which are enough for production of 3 times ZT-250.
- ❖ The budget that the company settled for the amount of all these materials is 2.390.700 ₺.

SOLUTION METHODOLOGY

Materials that are used in the production of the drilling machine ZT250:

	Materials	Number of pieces in one ZT250	€/pieces	Total volume (m^3)	Total Price (€)	Delay time if they are late supplied (day)
				$v_{s,i}$	$p_{s,i}$	$t_{s,i}$
a	Electrical and Electronic Components					
<i>a1</i>	Sensors	20	50	0,03	1000	0,2
<i>a2</i>	Relays	50	20	0,01	1000	0,2
<i>a3</i>	Capacitors	100	10	0,07	1000	0,2
<i>a4</i>	Transformers	5	1.000	0,01	5000	0,5
<i>a5</i>	Power sources	2	5.000	0,06	10000	0,5
<i>a6</i>	Control units	1	10.000	0,80	10000	1
b	Metal and Plastic Parts					
<i>b1</i>	Propellers	2	4.000	0,30	8000	0,5
<i>b2</i>	Gears	50	50	0,03	2500	1
<i>b3</i>	Fasteners	800	3	0,08	2400	0,5
c	Coating Materials					
<i>c1</i>	Paint (1L)	20	200	0,3	4000	0,5
<i>c2</i>	Varnish (1L)	10	120	0,5	1200	1
<i>c3</i>	Anticorrosive coatings (1 m ²)	50	500	1,5	25000	3
d	Lubrication and Cooling Materials					
<i>d1</i>	Oil (1L)	100	80	1,5	8000	2
<i>d2</i>	Grease (1kg)	50	100	0,1	5000	2
<i>d3</i>	Coolant (antifreeze) (1L)	50	100	0,2	5000	2
e	square profiles					
<i>e1</i>	200x200mm	2	1500	0,5	3000	4
<i>e2</i>	120x120mm	1	1200	0,1	1200	4
<i>e3</i>	100x100mm	8	900	0,5	7200	4
<i>e4</i>	Lama steel (Cold Drawing)	8	1000	2,5	8000	4
f	Bearings					
<i>f1</i>	Bearing (A30204)	24	250	0,2	6000	4
<i>f2</i>	Bearing (C36028)	12	150	0,1	1800	4

g	Iron sheets					
<i>g1</i>	15mm Iron Sheet	250	210	5	52500	7
<i>g2</i>	10mm Iron Sheet	200	210	4	42000	6
h	Main parts					
<i>h1</i>	Cummins F6L913 Engine	1	100.000	1,8	100000	14
<i>h2</i>	Tsubaki chain (1kg)	100	3.000	0,6	300000	10
<i>h3</i>	Parker rotation engine	1	8000	0,1	8000	14
<i>h4</i>	Rexroth A6VM80 crane	1	30.000	2	30000	14
<i>h5</i>	Parker Hydraulic Pumps	2	10.000	0,08	20000	7
<i>h6</i>	Triplex 135 Mud Pump	1	50.000	0,2	50000	7
j	Mounting Parts					
<i>j1</i>	Loaves	250	2	0,02	500	0,5
<i>j2</i>	Bolts	300	1,00	0,02	300	0,5
<i>j3</i>	Screws	350	2	0,02	700	1
<i>j4</i>	Washers	500	2	0,02	1000	1
k	Accessories					
<i>k1</i>	Delhi pipe (1m)	100	100	2	10000	2,5
<i>k2</i>	Drill bits	800	20	3	16000	5
<i>k3</i>	Shock absorbers	50	250	1,8	12500	3
l	Spare Parts					
<i>l1</i>	Engine parts	25	600	1,6	15000	3
<i>l2</i>	Bearings	15	80	0,06	1200	2
<i>l3</i>	Gears	50	140	0,8	7000	1,5

TABLE 1: Materials

Sum of total prices: ₺783.000,00

Total volumes: 32,51 m³

The number of pieces which is used in the production of one ZT250 is considered a package.

Discount	$dscnt_{s,i}$ Discount percentage	$limit_{s,i}$ should be purchased (package)
<i>a.2</i>	0.2	6
<i>a.5</i>	0.15	6
<i>h.3</i>	0.25	6
<i>h.5</i>	0.1	5
<i>h.6</i>	0.1	5

TABLE 2: Discount information of particular materials

ERD Drilling Machines wants to buy at least 3 of these materials and stock it for 3 months. Most clear demand of company is that they will stock these materials for 3 months. Since the total volume of materials is $3 * 32,51 = 97,53$, we will have to supply some of the materials later and wait in the process for their arrival. This waiting time will be considered as an opportunity cost.

ERD Drilling Machines sells one ZT 250 Q8 averagely 1.200.000 ₺.

Our cost at first in our model which means total budget is 2.390.700 ₺ for 3 months. For one month it is 783.000 ₺. The company would be able to produce one ZT 250 in one month if they don't have space constraints etc. So, we will think $(1.200.000 - 783.000)/30 = 13.900$ as daily profit. It is not exact profit since the company has another cost, but it is the case in our simplified model as we don't have enough access all the data of the company.

Since the daily profit of the company is 13.900 ₺, the total waiting time will be considered as new 13.900 cost per day because it extends the procedure and prevent the company to get this profit.

Sets

s working areas / a, b, c, d, e, f, g, h, j, k, l /

i required materials / 1, 2, 3, 4, 5, 6 /

Parameters

$v_{s,i}$: volume(space) possessed by i materials under the s working area (m^3)

$t_{s,i}$: waiting time for the missing material i under the working area of s (day)

$p_{s,i}$: price of one package of i material under the s working area (₺)

$dscnt_{s,i}$: discount percentage for some i materials under s working areas

$limit_{s,i}$: threshold number of purchases which should be reached in order to get discount.

Scalar

u : daily profit received from the production of ZT-250 (₺) = 13900

y : total space for the stocking of materials (m^3) = 80

q : total delay cost per volume - we will check if discounts cover this cost = 18735

r : total budget for at least 3 times of each material = 2 390 700

Decision Variables

$x_{s,i}$: number of purchases of packages of i materials under the s working area

$down_{s,i}$: How many packages supplied less than 3 and will be supplied on the process and caused delay due to the insufficient space.

$up_{s,i}$: How many packages supplied more than 3 due to the discount.

mm : How much does this waiting cost company.

dd : How many days will be waited for the supplying of materials which will be purchased later.

ss : what is the sum of space of these late materials (m^3)

Binary Variable

$w_{s,i}$ if $p_{s,i} > 1$ =1, 0, otherwise. (Just for making sure that invalid si indices take the same value.)

Objective Function

$$\min Z = \sum_{s=a}^l \sum_{i=1}^6 down_{s,i} * t_{s,i} * u + (x_{s,i} + down_{s,i}) * p_{s,i} - x_{s,i} * dscnt_{s,i} * p_{s,i} - up_{s,i} * p_{s,i} \quad (1)$$

Constraints

$$x_{s,i} + down_{s,i} - up_{s,i} = 3 * w_{s,i} \quad (2)$$

$$\text{if } \frac{x_{s,i} * dscnt_{s,i} * p_{s,i}}{x_{s,i} * v_{s,i}} > \frac{mm}{ss} : x_{s,i} = limit_{s,i} \quad (3)$$

$$\sum_{s=a}^l \sum_{i=1}^6 x_{s,i} * v_{s,i} \leq \gamma \quad (4)$$

$$\sum_{s=a}^l \sum_{i=1}^6 x_{s,i} * down_{s,i} * p_{s,i} - x_{s,i} * dscnt_{s,i} * p_{s,i} - up_{s,i} * p_{s,i} \leq r \quad (5)$$

$$\sum_{s=a}^l \sum_{i=1}^6 \text{down}_{s,i} * t_{s,i} * u = \text{mm} \quad (6)$$

$$\sum_{s=a}^l \sum_{i=1}^6 \text{down}_{s,i} * t_{s,i} = \text{dd} \quad (7)$$

$$\sum_{s=a}^l \sum_{i=1}^6 \text{down}_{s,i} * v_{s,i} = \text{ss} \quad (8)$$

$$x_{s,i}, \text{down}_{s,i}, \text{up}_{s,i}, \text{mm}, \text{dd}, \text{ss} \geq 0, \forall i$$

(1)

First written cost is the total opportunity cost due to the delay in process, second is total price for purchases, third is the total discount (£) we should subtract it from total cost, fourth step is about amounts of packages which are purchased more than 3 times. Consequently, the cost of $\text{up}_{s,i}$ should be belonged to next period's budget.

(2)

Penalty and discounts constraints.

(3)

When the model separates out 1 m^3 , it settles with its delay time cost. In this model, it is mm/ss. We have found and entered this number = q = 18735. It means that if get more discount than this number per 1 m^3 , the model will accept that number since it is more profitable.

(4)

Space constraint

(5)

Budget constraint. $up_{s,i} * p_{s,i}$ also was excluded here because it belongs to next terms' budget.

(6), (7) and (8) were explained in the “variables” part.

RESULTS

Yes, we have reached an optimal solution.

Numerical Results

$x_{s,i}$

The list of amounts of packages of materials which the company should purchase.

s/i	1	2	3	4	5	6
a	3	6	2	3	6	3
b		3	3			
c	3	3	3			
d		3	3			
e	3	3	3	3		
f	3	3				
g	2	3				
h	3	3	6	3	5	5
j	3	3	3	3		
k		3	3			
l	3	3	3			

$down_{s,i}$

The list of amounts of packages of materials which the company should not supply at first and wait for them in the process.

s/i	1	3
a		1
b	3	
d	3	
g	1	
k	3	

$up_{s,i}$

The list of amounts of packages of materials which the company should purchase and stock for the next terms due to the discounts.

s/i	2	3	5	6
a	3		3	
h		3	2	2

Total cost = **2.600.380 ₺**

mm : How much does these waiting for extra supply cost = **308.580 ₺**

dd : How many days will be waited for the suppling of materials which will be purchased lately) = **22.200 days**

ss : What is the sum of space of these late materials (m^3) = **16.470 m^3**

Analysis of Solution

Our model is for 3 months (3 times ZT250)

At the moment the company is receiving a 2-weeks extension for 4 weeks of planned production of Zt-250.

Company's time extension: $3 \times 7 \times 2 = 42$ days

Our time extension= 22.2 days

Improvement: 19.8 days

Company's current total cost: $2,390,700 + 42 \times 13.9 = 2.974.500$ ₺

Our total cost: 2.600.380 ₺

Improvement: 374.000 ₺

CONCLUSION

In conclusion, we have learned that the company produces mainly two drilling machines and one of them, the ZT250 drilling machine, sells more and provides more profit. Presently, the company faces problems in the completion of the production of ZT250 drilling machines at the scheduled time due to the insufficient planning of production. We have taken the data and used this data in our mathematical model as reasonable parts. Exact numbers were quite difficult, but we have tried to achieve the most convenient mathematical model and to reach the optimal solution. We have achieved 19.8 days and Improvement and 374.000 ₺ improvements.

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